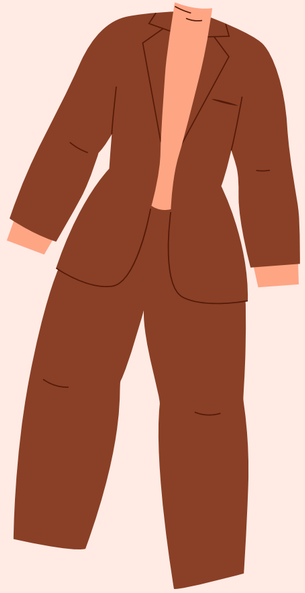


Weather Based Clothing Recommendation System

London Gibson-Purcell DSC 345

INTRODUCTION

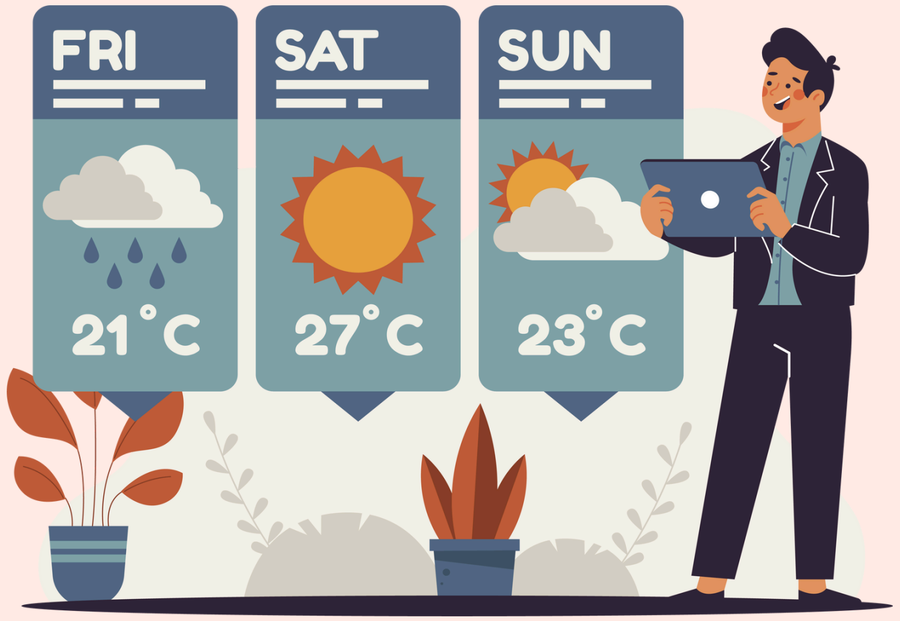
- Using AI to assist with daily decision-making through clothing recommendations

- A model system that suggests the most appropriate clothing category, along with customized choices based on closet

Business Problem

- People usually spend time and effort trying to decide what to wear everyday

- Opportunity for AI to use personalized data-driven recommendations to alleviate the hassle for consumers



SMART OBJECTIVES

Specific: Implement an AI model that uses weather data to suggest a clothing recommendation

Achievable: Using raw weather data, rule-based label creation, and user closet data

Measurable: Achieve 90% in predicting the correct clothing category using a decision-tree classifier

Relevant: helps users decide what to wear optimizing time

Success Metrics: model accuracy (precision & recall), clear interpretability, integration of app prototype, overall user satisfaction

Feasibility & Competition

Data

Clean Weather Data Readily Available
User Wardrobe Input

Model Creation

Clothing categories generated using rule-based logic
Simple tagging
Decision tree classifier - lightweight & interpretable

Simple Recommendation Pipeline: **Weather → Clothing Category → User Clothing Match**

Competition/Scalability

Little competition
Combines weather services & the retail industry
Revenue opportunity/Low Implementation Cost

ROAD MAP

Personalization Layer

Advanced Features

Data Foundation

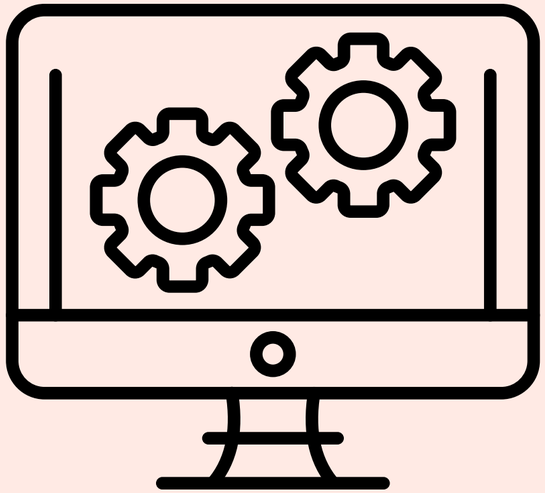
User Interface - App/Smart Home

Business Operation & Commercial Use

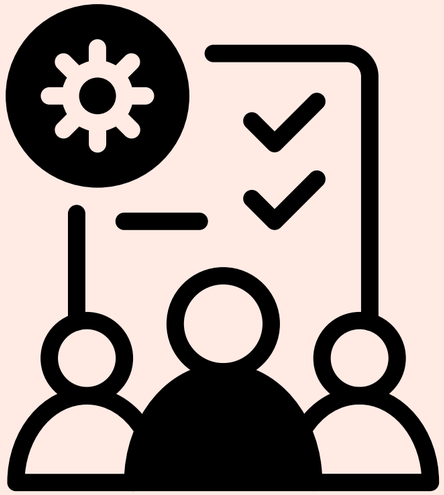
Team Assembly

Data Scientist/Analyst

Fashion Expert
Meterologist

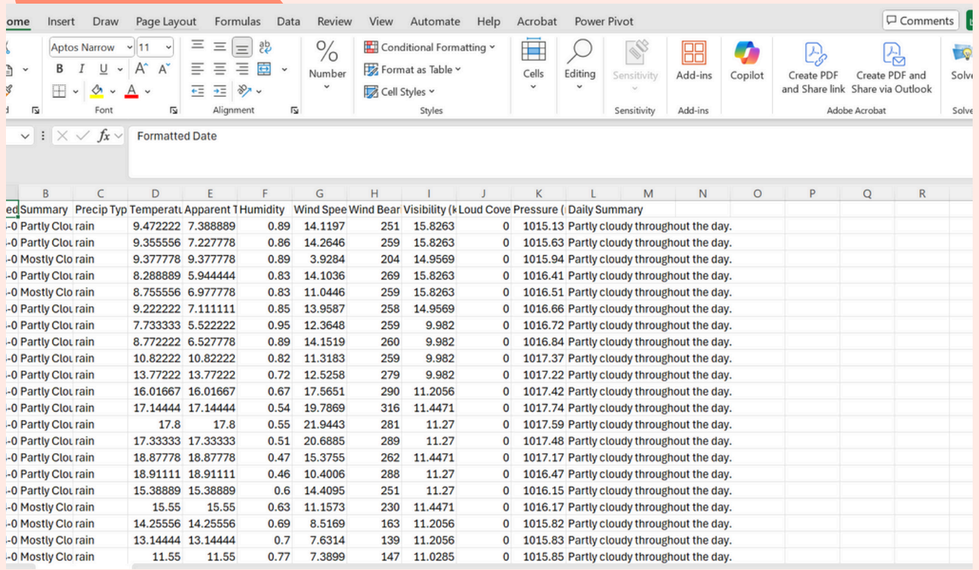


Backend Engineer/Mobile
Developer



Project Manager

Data Collection & Preparation



ed	Summary	Precip Typ	Temperatu	Apparent T	Humidity	Wind Spee	Wind Bear	Visibility (k	Loud Cove	Pressure (	Daily Summary
-0	Partly Clou	rain	9.472222	7.388889	0.89	14.1197	251	15.8263	0	1015.13	Partly cloudy throughout the day.
-0	Partly Clou	rain	9.355556	7.227778	0.86	14.2646	259	15.8263	0	1015.63	Partly cloudy throughout the day.
-0	Mostly Clo	rain	9.377778	9.377778	0.89	3.9284	204	14.9569	0	1015.94	Partly cloudy throughout the day.
-0	Partly Clou	rain	8.288889	5.944444	0.83	14.1036	269	15.8263	0	1016.41	Partly cloudy throughout the day.
-0	Mostly Clo	rain	8.755556	6.977778	0.83	11.0446	259	15.8263	0	1016.51	Partly cloudy throughout the day.
-0	Partly Clou	rain	9.222222	7.111111	0.85	13.9587	258	14.9569	0	1016.66	Partly cloudy throughout the day.
-0	Partly Clou	rain	7.733333	5.522222	0.95	12.3648	259	9.982	0	1016.72	Partly cloudy throughout the day.
-0	Partly Clou	rain	8.772222	6.527778	0.89	14.1519	260	9.982	0	1016.84	Partly cloudy throughout the day.
-0	Partly Clou	rain	10.82222	10.82222	0.82	11.3183	259	9.982	0	1017.37	Partly cloudy throughout the day.
-0	Partly Clou	rain	13.77222	13.77222	0.72	12.5258	279	9.982	0	1017.22	Partly cloudy throughout the day.
-0	Partly Clou	rain	16.01667	16.01667	0.67	17.5651	290	11.2056	0	1017.42	Partly cloudy throughout the day.
-0	Partly Clou	rain	17.14444	17.14444	0.54	19.7869	316	11.4471	0	1017.74	Partly cloudy throughout the day.
-0	Partly Clou	rain	17.8	17.8	0.55	21.9443	281	11.27	0	1017.59	Partly cloudy throughout the day.
-0	Partly Clou	rain	17.33333	17.33333	0.51	20.6885	289	11.27	0	1017.48	Partly cloudy throughout the day.
-0	Partly Clou	rain	18.87778	18.87778	0.47	15.3755	262	11.4471	0	1017.17	Partly cloudy throughout the day.
-0	Partly Clou	rain	18.91111	18.91111	0.46	10.4006	288	11.27	0	1016.47	Partly cloudy throughout the day.
-0	Partly Clou	rain	15.38889	15.38889	0.6	14.4095	251	11.27	0	1016.15	Partly cloudy throughout the day.
-0	Mostly Clo	rain	15.55	15.55	0.63	11.1573	230	11.4471	0	1016.17	Partly cloudy throughout the day.
-0	Mostly Clo	rain	14.25556	14.25556	0.69	8.5169	163	11.2056	0	1015.82	Partly cloudy throughout the day.
-0	Mostly Clo	rain	13.14444	13.14444	0.7	7.6314	139	11.2056	0	1015.83	Partly cloudy throughout the day.
-0	Mostly Clo	rain	11.55	11.55	0.77	7.3899	147	11.0285	0	1015.85	Partly cloudy throughout the day.

Weather Dataset- over 9500 rows from 2006-2016

Business Implementation- partnership with & access to official weather collection data from NWS/Apple/Google

User uploads wardrobe through text or images (computer vision)

Items are tagged by type (jacket, shirt, jeans, etc.), warmth level, weather suitability

Data Collection & Preparation Pt.2

Rule-based Logic is used to determine the base clothing layer with temperature; suggestions are modified based on weather conditions

```python
•[24]: def get_clothing_label(row):
    temp = row['temp_f']
    rain = row['is_rain']
    snow = row['is_snow']
    wind = row['is_windy']


    if temp < 40:
        label = "Heavy Coat"
    elif temp < 60:
        label = "Jacket"
    elif temp < 75:
        label = "Long Sleeve"
    else:
        label = "T-shirt"


    if snow == 1:
        label = "Heavy Coat + Boots"
    if rain == 1:
        label += " + Raincoat"
    if wind == 1:
        label += " + Windbreaker"

    return label

df['label'] = df.apply(get_clothing_label, axis=1)
print(df[['temp_f', 'is_rain', 'is_snow', 'is_windy', 'label']])
```

Weather Data (Temp, Rain,Snow, Wind)

↓

ML Model Prediction

→ "Jacket + Raincoat"

↓

Digital Closet Filtering

(find users' jackets & raincoats)

↓

Personalized Recommendations

- Yellow Raincoat (best rain match)
- Blue Denim Jacket (backup option)

CLOSET MATCHING ENGINE

Layer 1: Category Matching

Take the predicted clothing label treated as a substring and match it to the closet through substring or token matching

Layer 2: Modifier Extraction

The modifiers based on the weather conditions become requirements that the closet must satisfy

Layer 3: Scoring System

Rank clothing items that are best suitable for the combination of conditions (category, modifier, environmental suitability, user comfort, past usage of item etc.)

TOOLS & PACKAGES

Pandas

matplotlib

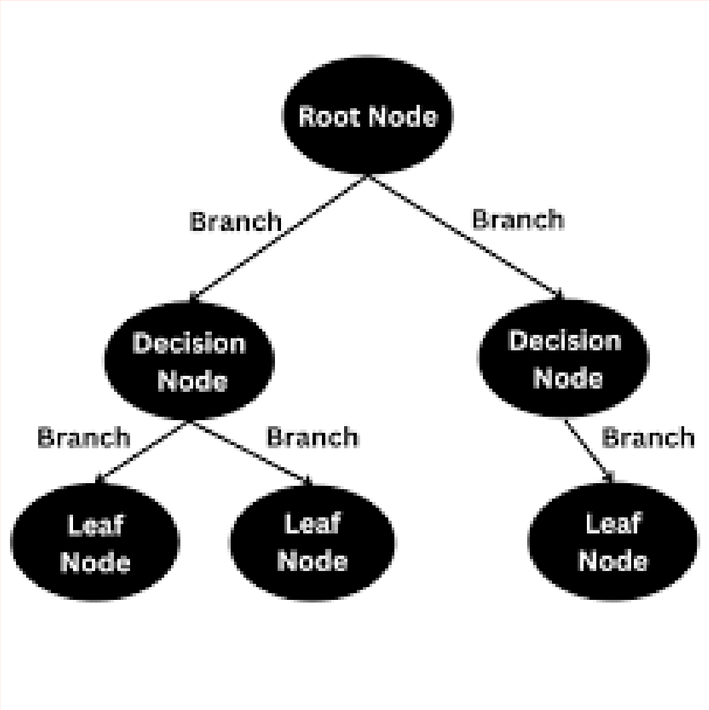
NumPy

PyTorch

scikit learn

OpenWeather

MODEL SELECTION



Logistic Regression - too simple to handle multiple conditions; clothing decisions aren't linear

Random Forest - multiple decision trees, not best for basic logic understanding, heavier computation, better for later models

Decision Tree Classifier - easy to interpret, follows if/then logic, can handle personalization changes

Generalization

train 80% data- model will learn the patterns & logic

test 20% data - evaluate new input

```
6]: from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
df['label_encoded'] = le.fit_transform(df['label'])

x= df[['temp_f', 'Humidity', 'Wind Speed (km/h)', 'is_rain', 'is_snow', 'is_windy']]
y = df['label_encoded']
```

```
7]: from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

```
8]: from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier(max_depth=5, random_state=42)
model.fit(X_train, y_train)
```

```
8]: DecisionTreeClassifier
DecisionTreeClassifier(max_depth=5, random_state=42)
```

<u>Precision-</u> How often is the model predicting the right clothing category?

<u>Recall-</u> How well does the model capture all correct clothing situations?

OVERFITTING

Reliable for New User Input

Consistent Recommendations

Works across different conditions

1) Limit Model Capacity: ensure that the training phase data and patterns don't get too complicated

2) Rule-Based Label Generation: straightforward labels will make the model less sensitive to noise (difficult to mess up)

3) Consistently evaluate new data during the training phase

MiNiMuM ViaBLe PRODuCT

1) Manual weather input
2) Decision Tree predicts clothing category
3) Digital closet stores items
4) Matching engine maps prediction → closet items
5) Displays personalized outfit recommendations

```
[35]: matches = recommend_from_closet(pred_label, digital_closet)

print("Predicted Clothing Label:", pred_label)
print("\nMatching Closet Items:")
for item in matches:
    print("-", item["name"], "(", item["category"], ")")
```

```
Predicted Clothing Label: Jacket + Raincoat

Matching Closet Items:
- Blue Denim Jacket ( Jacket )
- Beige Trench Coat ( Jacket )
- Yellow Raincoat ( Raincoat )
```

```
[32]: digital_closet = [
    {"name": "Black Puffer Coat", "category": "Heavy Coat"},
    {"name": "Blue Denim Jacket", "category": "Jacket"},
    {"name": "Beige Trench Coat", "category": "Jacket"},
    {"name": "Gray Hoodie", "category": "Long Sleeve"},
    {"name": "White T-shirt", "category": "T-shirt"},
    {"name": "Yellow Raincoat", "category": "Raincoat"},
    {"name": "Snow Boots", "category": "Boots"},
    {"name": "Blue Windbreaker", "category": "Windbreaker"}
]
```

```
[33]: def recommend_from_closet(predicted_label, closet):
    predicted_label = predicted_label.lower()
    matches = []

    for item in closet:
        cat = item["category"].lower()
        if cat in predicted_label:
            matches.append(item)

    return matches
```

```
[34]: sample_weather = X_test.iloc[0:1]
pred_encoded = model.predict(sample_weather)[0]
pred_label = le.inverse_transform([pred_encoded])[0]
pred_label
```

```
[34]: 'Jacket + Raincoat'
```

DEPLOYMENT

Backend

- Expose model through an API making it more accessible
- def api_recommend(weather_input, closet_items)
- return {"prediction": pred_label, "recommendations": matches}

Frontend

Possible Future Scaling

- Host API on cloud service platform
- store user closet data in real real-time database
- Integrate live weather API

LIMITATIONS

1 Data: doesn’t account for unusual weather events & severity, clothing labels are rule-based

2 Labeling Model: Decision Tree oversimplifies complex weather–clothing relationships and does not learn user-specific comfort levels or behaviors

3 Matching Engine: layering or nuanced outfit logic is not implemented (basic implementation)

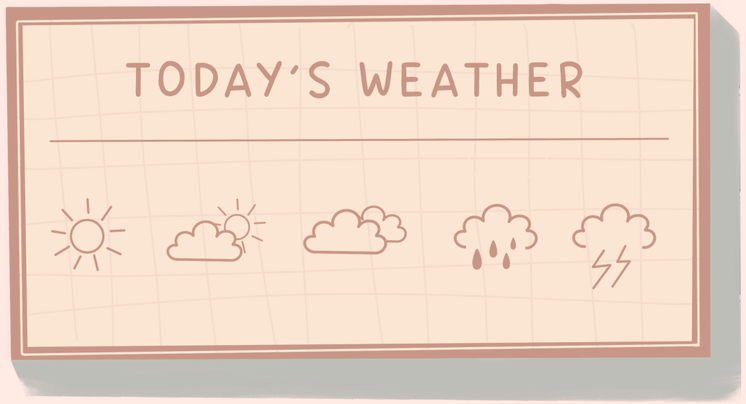
ETHICAL CONSIDERATIONS

Closet Data Privacy

Fairness/Inclusivity

Transparency

User Control



FUTURE WORK

Advanced Models

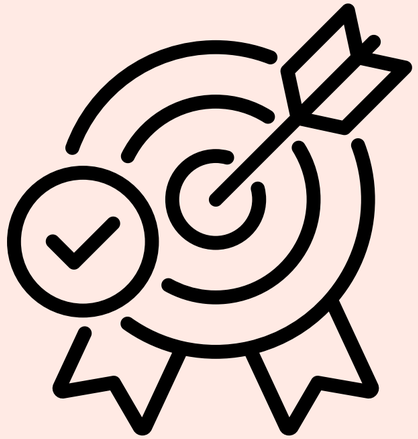
Personalization

Closet Automation

Full App Deployment

CONCLUSION

Weather data + AI + a user's digital closet = personalized outfit recommendations



MVP with prediction matching engine clear pipeline

save time & maximize use of their wardrobe

Strong foundation for real-world applications; mobile app, smart home feature